

**1** It is given that  $y = \sinh(x^2) + \cosh(x^2)$ .

- (a) Use standard results from the list of formulae (MF19) to find the Maclaurin's series for  $y$  in terms of  $x$  up to and including the term in  $x^4$ . [2]

[illegible]

- (b) Deduce the value of  $\frac{d^4y}{dx^4}$  when  $x = 0$ . [1]

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- (c) Use your answer to part (a) to find an approximation to  $\int_0^{\frac{1}{2}} y \, dx$ , giving your answer as a rational fraction in its lowest terms. [2]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



- (b)** It is given instead that  $a = b = 0$ .

Find the value of  $\frac{d^2y}{dx^2}$  when  $t = 1$ .

[4]

This image shows a full page of a document template designed for handwriting practice or general writing. It consists of approximately 28 evenly spaced horizontal dotted lines running across the width of the page. The background is plain white, and there are no margins, headers, footers, or other markings present.